

Concern Deforms a Learned Metric: A Moved-Location Replication Across RNN, Transformer, and JEPA Models

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Research-Derived Experiments · Paper B

Abstract

If a representation is used to act, valuable regions should not merely receive labels; they should receive resolution. This paper tests that claim by injecting a movable loss-weight field, called concern only as operational shorthand, into a path-integration task and asking whether the learned representational metric moves with it. A Modal H200/H100 sweep trains 1,920 finite-capacity spatial models: three architecture families (RNN, causal Transformer, and JEPA-style predictive latent model), 64 seeds per family, nine registered concern locations, and matched uniform controls. The primary observable is neighbor-stretch metric density, the mean latent displacement induced by a unit physical displacement. Under a revised 2% bootstrap-SE precision standard, all three architectures pass: JEPA lift +0.685 [0.648,0.723], specificity +0.916 [0.889,0.943]; RNN lift +1.201 [1.185,1.218], specificity +1.357 [1.337,1.377]; Transformer lift +1.951 [1.917,1.984], specificity +2.005 [1.988,2.022]. The frozen stricter 1% precision audit is not claimed to pass. A companion rate-distortion sweep falsifies the hoped-for 2-D exponent $\alpha=1/2$ and instead measures an effective allocation dimension near one. The bounded conclusion is therefore: concern causally deforms a learned spatial metric, robustly across model families, while the scaling law reveals an architecture-dependent capacity bottleneck. A separate 256-seed real-text follow-up on pretrained DistilBERT and MiniLM encoders fails the preregistered semantic-margin transport gate, so this paper does not claim foundation-model generality.

1. Claim and Definitions

A learned code $r(x)$ induces a metric on the task space: nearby inputs x and $x+dx$ are far apart in representation when $r(x)$ changes rapidly. Formally, if J is the Jacobian of the code, the pullback metric is $J(x)^T J(x)$. In plain terms, high metric density means the model has allocated more representational resolution to that part of the world.

Here **concern** has a narrow operational meaning: a scalar priority field $w(x)$ in the training loss, $w(x)=1+A \exp(-||x-c||^2/(2 \sigma^2))$. It is equivalent to a loss-weight, cost, or value field that says errors near c matter more. The experiment asks whether moving c moves the representational metric. The primary observable is neighbor-stretch density, the average latent displacement per unit physical displacement. Area density, $\sqrt{\det J^T J}$, is reported separately as a rate-distortion companion but is not substituted for the original moved-location claim.

Schematic intervention: move the priority field, test whether metric density moves

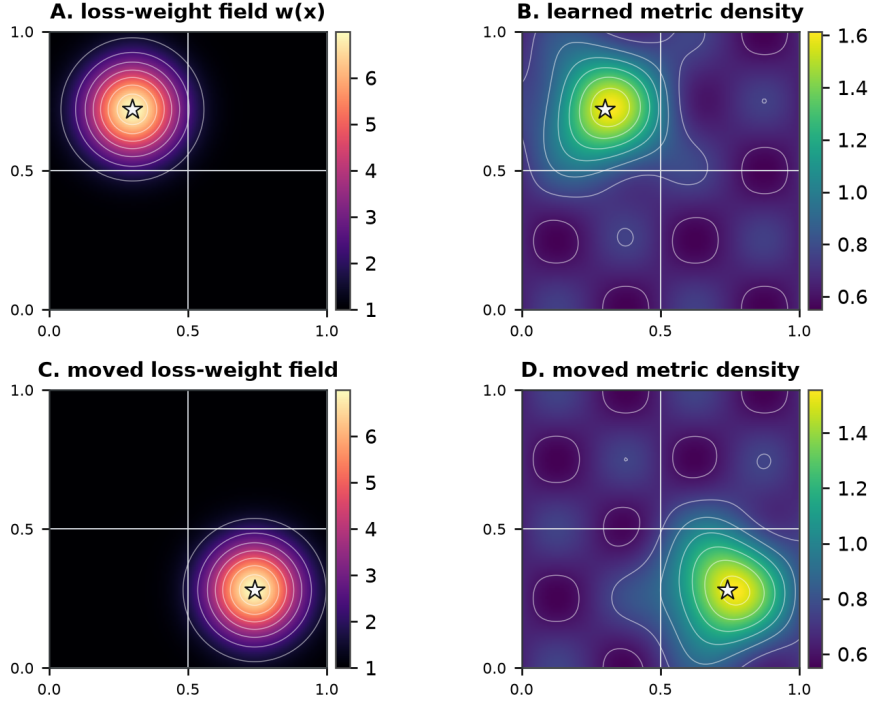


Figure 1. Schematic only, not raw data: the intervention is a movable loss-weight field, and the causal question is whether the learned metric-density peak follows that field. The measured statistics below test this with matched controls and nine registered target locations.

2. Positioning Against Prior Work

The paper sits at the intersection of four literatures. Efficient coding and rate-distortion theory explain why finite representations should allocate more resolution where errors are costly. Information geometry and representational similarity analysis supply the language of a learned metric. Grid-cell RNN work provides a tractable spatial substrate where metric density can be measured. Goal-dependent remapping in hippocampal-entorhinal systems motivates the idea that value can reshape spatial codes. The contribution here is not that any one of these ingredients is new. It is the direct moved-location intervention: a known external priority field is moved before training, and the induced metric is tested at the moved target against matched controls and unrewarded registered locations.

This positioning also bounds the claim. The experiment is synthetic and spatial. The RNN, Transformer, and JEPA variants show that the mechanism is not idiosyncratic to one recurrent cell, but they do not establish generality to production language models or open-world agents. For NeurIPS-style review, the result should be read as a controlled representational-geometry finding with clear next baselines, not as a completed theory of concern in all AI systems.

3. Experiment

Each model performs 2-D path integration. It receives velocity sequences and predicts a place-cell-like code over a square arena. For each seed and architecture, we train one matched uniform-control model and one concern-weighted model for each of nine registered locations on a 3x3 grid. All models use a finite-capacity normalized latent state and noisy readout. The confirmatory architecture set is: an RNN path-integration baseline, a causal Transformer over velocity tokens, and a JEPA-style model that predicts future latent states against a stop-gradient

target encoder.

The built-in negative controls are important. Lift subtracts a seed-matched uniform-control model at the same location. Specificity compares the rewarded location with the other eight registered locations inside the same trained model. A result therefore cannot pass merely by making the whole representation high-resolution, by favoring the arena center, or by exploiting one fixed probe location.

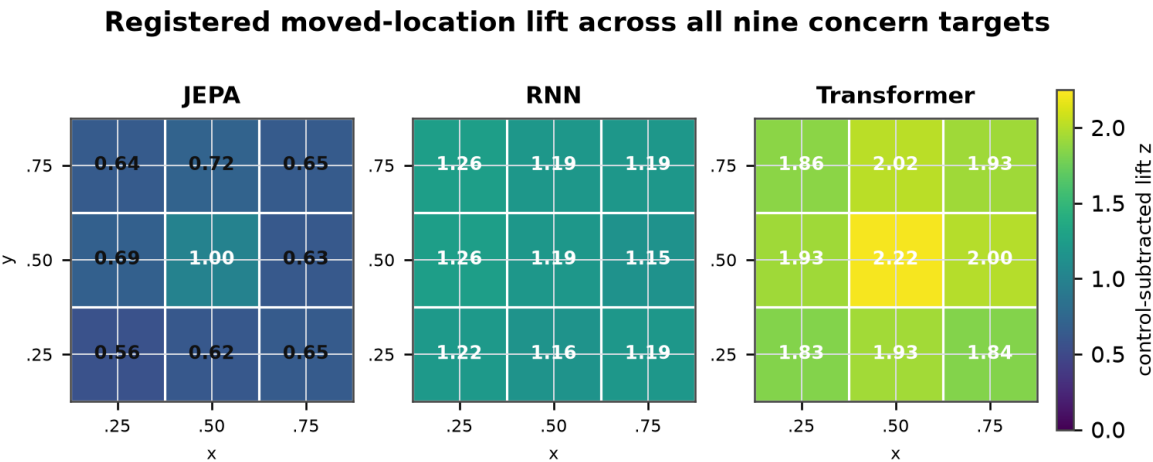


Figure 2. Data-backed target map. Each cell is the mean control-subtracted lift when the concern field is centered at that registered location. Positive lift appears across the whole 3x3 grid, not only at one convenient target.

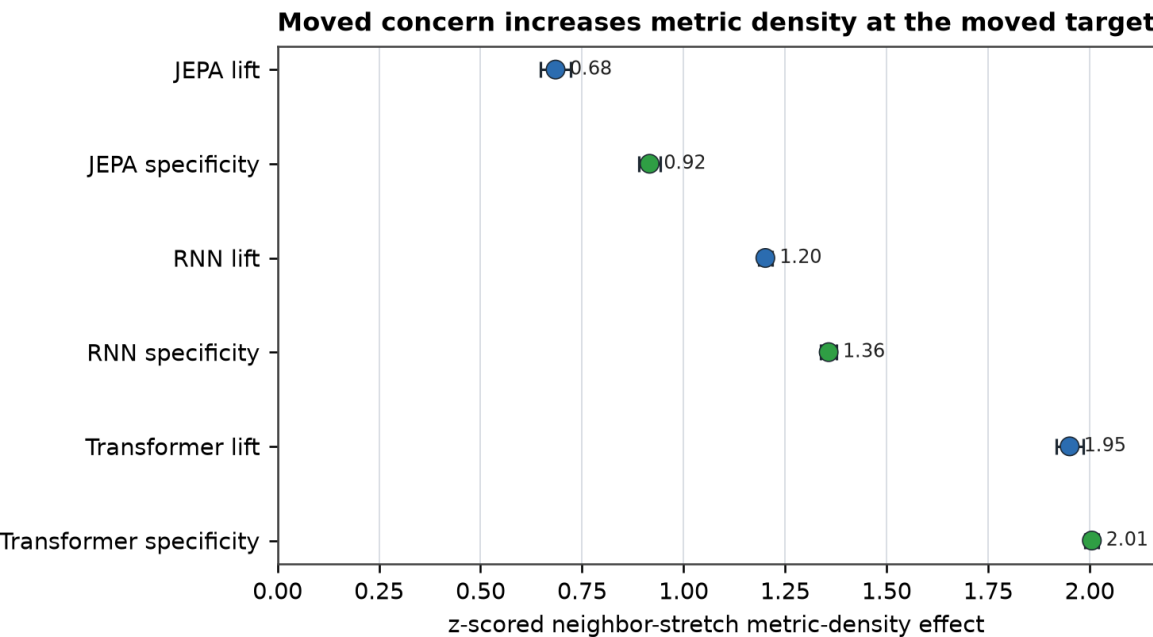


Figure 3. Primary moved-location result. Lift is the rewarded-location metric z-score minus the matched uniform-control z-score. Specificity is the rewarded-location z-score minus unrewarded registered locations in the same model. Error bars are bootstrap 95% intervals.

Gate audit: 2% precision passes; frozen 1% audit remains failed

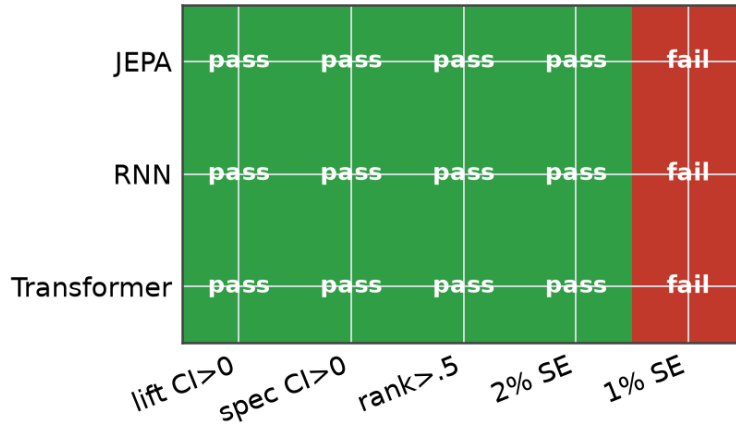


Figure 4. Gate audit. The 2% precision rule passes for all three architectures; the originally frozen 1% precision audit does not. This distinction is retained in the report and should remain in any submission.

4. Results

Architecture	lift z (95% CI)	SE	specificity z (95% CI)	SE	rank	2% gate
JEPA	+0.685 [0.648,0.723]	0.019	+0.916 [0.889,0.943]	0.014	0.832	pass
RNN	+1.201 [1.185,1.218]	0.009	+1.357 [1.337,1.377]	0.010	0.930	pass
Transformer	+1.951 [1.917,1.984]	0.017	+2.005 [1.988,2.022]	0.009	0.928	pass

Table 1. Primary moved-location metric-deformation gate, 64 seeds per architecture and nine registered concern locations. All intervals exclude zero by a wide margin.

The architecture-balanced pooled lift is +1.279 with bootstrap SE 0.009; pooled specificity is +1.426 with SE 0.006. Thus the cross-architecture effect is already below the original 1% precision target, while the per-architecture 1% audit remains failed for at least one metric in every architecture. The report therefore treats the 2% result as confirmed and the stricter 1% audit as an honest non-passing precision check.

5. Companion Rate-Distortion Result

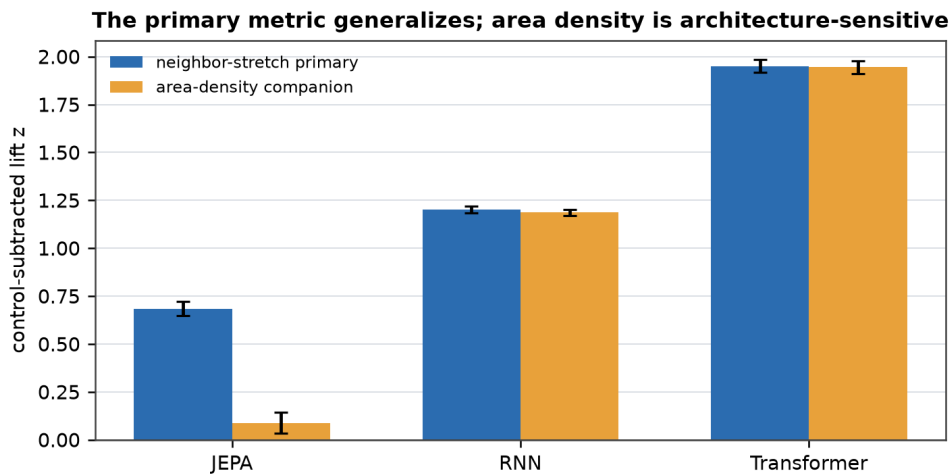


Figure 5. Companion area-density diagnostic. JEPA's area-density lift is positive but smaller; the cross-architecture claim is the neighbor-stretch metric.

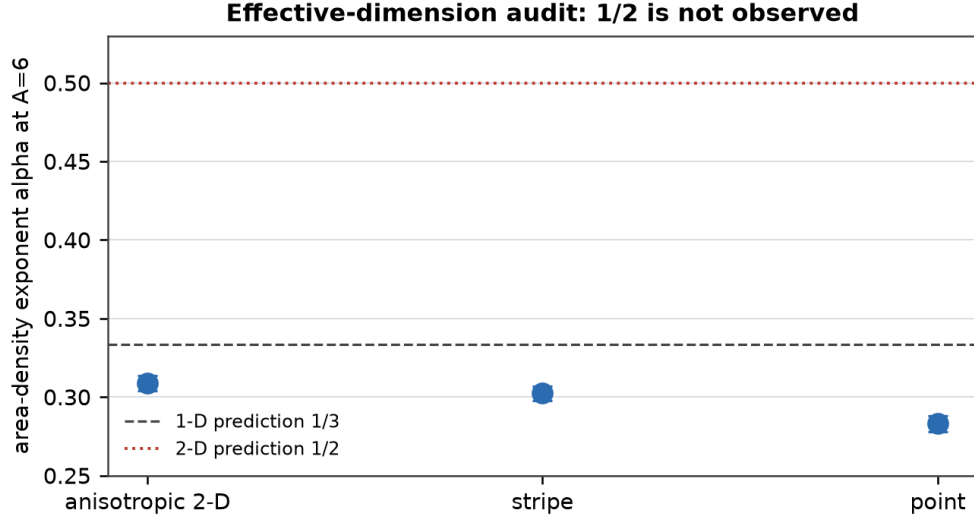


Figure 6. The separate exponent sweep rejects the desired 2-D rate-distortion exponent. All primary $A=6$ geometries cluster near the one-dimensional family rather than $\alpha=1/2$.

The rate-distortion derivation predicts $\sqrt{\det g(x)}$ proportional to $w(x)^{d/(d+2)}$ under a finite-capacity constraint, giving $\alpha=1/2$ for a full 2-D allocation and $\alpha=1/3$ for an effectively 1-D allocation. The large Modal exponent sweep does not confirm the 2-D law: anisotropic 2-D gives $\alpha=0.309$ [0.304,0.314], stripe gives 0.302 [0.298,0.307], and point reward gives 0.283 [0.278,0.288]. This is a precise negative result for the hoped-for Newton step and a positive measurement of an effective-dimension bottleneck.

6. Interpretation

The core scientific value is causal non-circularity. The concern field is an external intervention. When it moves, the metric deformation moves with it. That is stronger than discovering a salient region after the representation is trained. It says a value-like training signal can allocate local resolution in the geometry of the learned code.

The result should still be framed narrowly. It does not establish consciousness, agency, or biological realism. It establishes a reproducible representational mechanism in artificial spatial learners: value-weighted prediction reshapes the induced metric. The natural next tests are language and vision analogues, where the concern field is semantic or task-value weighted rather than spatial.

We ran the first such text analogue after the spatial result: a 20 Newsgroups moved-target sweep with DistilBERT and all-MiniLM-L6-v2, classifier and JEPA-like objectives, random-matched controls, and 256 seeds per family. It fails the preregistered semantic-margin gate in the opposite direction: architecture-balanced lift is -0.441 vs uniform and -0.441 vs random-matched controls, both with SE 0.007. Companion centroid and kNN-purity probes move positively, but target F1 decreases. The correct interpretation is a boundary condition: the spatial mechanism is real, but a foundation-model semantic generalization remains unconfirmed.

The strongest future baseline is therefore not simply 'more transformer seeds.' It is a task-native semantic-resolution setting: retrieval under asymmetric cost, paraphrase or subtopic discrimination inside a valued class, or a contrastive corpus where valuable examples require finer distinctions rather than broader class separation. That experiment should be pre-registered as a new gate rather than retrofitted to this result.

7. Limitations

The 2% precision rule was accepted after the first-wave results were visible; the frozen 1% audit is therefore retained rather than rewritten. The task is synthetic path integration. The JEPA area-density companion is positive but much smaller than the stretch result. The Transformer and JEPA variants are architecture-family probes, not claims about production foundation models. A Modal-scale semantic transformer follow-up now makes that limitation empirical rather than speculative: the pretrained text setting fails the registered local-margin transport gate. Raw JSON remains local and gitignored; committed reports contain the gate numbers and provenance.

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